Cloud Computing

DSC 204A: Scalable Data Systems

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Meme idea credit: https://datasystemsfun.tumblr.com/

"Above the Clouds": The Berkeley View (2009)

The paper that defined cloud computing — Armbrust, Fox, Patterson, Stoica, Zaharia, et al.

Cloud = SaaS + Utility Computing

Three things that were genuinely *new* in 2009:

1. **Illusion of infinite resources** on demand
2. **No up-front commitment** — start small, scale later
3. **Pay-as-you-go** at fine granularity

> *"1000 servers for one hour costs no more than 1 server for 1000 hours."* — elasticity at no premium

Why now? Large datacenters reached **5–7× cost advantage** in electricity, bandwidth, hardware, and ops via economies of scale.

Top 10 obstacles they predicted (still relevant!):

1. Availability of service
2. **Data lock-in**
3. Data confidentiality & auditability
4. Data transfer bottlenecks
5. **Performance unpredictability**
6. Scalable storage
7. Bugs in distributed systems
8. Scaling quickly
9. Reputation fate sharing
10. Software licensing

> 17 years later, **almost every one is still an active research / engineering challenge**.

Armbrust, Fox, Griffith, Joseph, Katz, Konwinski, Lee, Patterson, Rabkin, Stoica, Zaharia. Above the Clouds: A Berkeley View of Cloud Computing. UCB/EECS Technical Report 2009-28.

Ten Years Later: The Serverless Berkeley View (2019)

Same Berkeley team revisits the cloud — Jonas, Schleier-Smith, Sreekanti, Stoica, Patterson, et al.

Serverless = FaaS + BaaS

> *"Serverless cloud computing represents an evolution that parallels the transition from assembly language to high-level programming languages."*

The bold prediction:

> *"Serverless computing will grow to dominate the future of cloud computing."*

Why? Cloud 2.0 freed users from physical hardware, but **left them managing virtual hardware** — autoscaling, load balancing, patching, monitoring. Serverless removes this burden entirely.

5 limitations of today's serverless platforms:

1. **Inadequate storage** for fine-grained operations
2. **No fine-grained coordination** between functions
3. **Poor performance** for standard comm patterns (e.g., shuffle)
4. **Unpredictable performance** — cold starts, noisy neighbors
5. Limited memory (0.125–3 GiB) and runtime (≤900s)

Research challenges across:
abstraction · systems · networking · security · architecture

Concrete economics: AWS Lambda is ~**$0.0000002 / 100ms**; serverful EC2 is **$0.0000867–0.408 / 60s**. Different units, different mental models.

Jonas, Schleier-Smith, Sreekanti, Tsai, Khandelwal, Pu, Shankar, Carreira, Krauth, Yadwadkar, Gonzalez, Popa, Stoica, Patterson. Cloud Programming Simplified: A Berkeley View on Serverless Computing. UCB/EECS Technical Report 2019-3.

AWS Today: 240+ Services Across 30+ Categories

Compute – EC2, Lambda, ECS, EKS, Fargate, Batch, Lightsail, Elastic Beanstalk, App Runner, Outposts, Wavelength, Local Zones, EC2 Auto Scaling, EC2 Image Builder, Compute Optimizer, Snow Family

Database – RDS (MySQL / Postgres / MariaDB / Oracle / SQL Server), Aurora, DynamoDB, DAX, Redshift, ElastiCache (Redis / Memcached), MemoryDB, Neptune, DocumentDB, Keyspaces, Timestream, QLDB

Analytics – Athena, EMR, Redshift, Kinesis (Streams / Firehose / Analytics / Video), OpenSearch, QuickSight, Glue, Glue DataBrew, Lake Formation, MSK (Kafka), Data Pipeline, FinSpace, DataZone, Clean Rooms

App Integration – SQS, SNS, EventBridge, Step Functions, AppSync, MQ, SWF, MWAA (Airflow), AppFlow

Developer Tools – CodeCommit, CodeBuild, CodeDeploy, CodePipeline, CodeStar, Cloud9, X-Ray, CodeArtifact, CodeGuru, CodeWhisperer, CDK, Application Composer, FIS

Security & Identity – IAM, IAM Identity Center, Cognito, KMS, CloudHSM, Secrets Manager, GuardDuty, Inspector, Macie, WAF, Shield, ACM, Detective, Security Hub, Audit Manager, Network Firewall, Firewall Manager, Verified Access, Verified Permissions, RAM

Migration & Transfer – Migration Hub, Application Migration Service (MGN), DMS, DataSync, Snow Family (Snowball / Snowmobile / Snowcone), Transfer Family, Application Discovery Service

Media Services – MediaConvert, MediaLive, MediaPackage, MediaStore, MediaTailor, Elemental, IVS, Nimble Studio, Elastic Transcoder

Cost Management – Cost Explorer, Budgets, Cost & Usage Report, Pricing Calculator, Marketplace, Savings Plans

Storage – S3, S3 Glacier, EBS, EFS, FSx (Lustre / Windows / NetApp ONTAP / OpenZFS), Storage Gateway, Backup, DataSync, Elastic Disaster Recovery, Transfer Family

Networking & CDN – VPC, CloudFront, Route 53, API Gateway, ELB (ALB / NLB / CLB / GWLB), Direct Connect, Transit Gateway, Global Accelerator, App Mesh, PrivateLink, Cloud Map, VPC Lattice, Network Firewall

ML / AI – SageMaker, Bedrock, Q, Rekognition, Comprehend, Translate, Polly, Transcribe, Lex, Forecast, Personalize, Textract, Kendra, CodeWhisperer, HealthLake, A2I, DeepRacer, Panorama, Fraud Detector, Lookout

Containers – ECS, EKS, Fargate, ECR, Copilot, App2Container, Proton

Management & Governance – CloudWatch, CloudTrail, Config, Systems Manager, OpsWorks, CloudFormation, Service Catalog, Trusted Advisor, Control Tower, Organizations, License Manager, Resilience Hub, Launch Wizard

IoT – IoT Core, Greengrass, IoT Analytics, IoT Device Defender, IoT Events, IoT SiteWise, IoT TwinMaker, FreeRTOS, IoT Device Management, IoT FleetWise, IoT 1-Click

Mobile / Front-End – Amplify, AppSync, Pinpoint, Device Farm, Location Service

Business Apps – WorkSpaces, WorkDocs, Chime, Chime SDK, Connect, AppStream, WorkMail, Honeycode, Wickr, SES

Specialty – Braket (quantum), Ground Station (satellite), RoboMaker (robotics), Managed Blockchain, GameLift, Open 3D Engine

Source: aws.amazon.com/products – list updated continuously. Representative services shown per category. AWS publishes new services monthly.

The Big Six: Most Popular AWS Services

EC2 — Compute

Virtual servers in the cloud. The original AWS service (2006). Pay per second.

If you've used "the cloud," you've probably used EC2.

S3 — Object Storage

Buckets of objects, virtually unlimited. 11 9's of durability. The internet's storage layer.

Stores 280+ trillion objects (2023).

Lambda — FaaS

Run code without provisioning servers. Pay per 100ms of execution. Triggered by events.

10 trillion+ invocations / month.

RDS — Managed SQL DB

Postgres, MySQL, Aurora, etc. Automated backups, patching, replication. The "boring" workhorse.

Aurora: 5x faster than vanilla MySQL.

DynamoDB — NoSQL

Key-value & document, single-digit ms latency at any scale. Built originally to power Amazon.com.

89M req/s peak on Prime Day.

CloudFront — CDN

Edge cache + delivery network. 600+ POPs worldwide. Pairs with S3 for static sites.

Powers Disney+, Twitch, Hulu.

> **80% of AWS customers use these six.** Master them, and the rest is mostly variations and integrations.

More Heavy Hitters You Should Know

VPC — Virtual Network
Your private network in the cloud. Foundation for everything that runs in EC2.

IAM — Identity & Access
Who can do what. Roles, policies, federation. Mandatory for every account.

Route 53 — DNS
Authoritative DNS + traffic policies + health checks. Named after port 53.

API Gateway
REST/HTTP/WebSocket front door. Pairs with Lambda for serverless APIs.

SQS — Queue
Decouple producers from consumers. Standard or FIFO. Trillions of messages/day.

SNS — Pub/Sub
Push notifications, fan-out to email, SMS, Lambda, SQS. Multi-protocol.

EKS / ECS — Containers
Managed Kubernetes (EKS) or AWS-native ECS. Pair with Fargate for serverless containers.

CloudWatch — Observability
Metrics, logs, alarms, dashboards. The eyes and ears of every AWS workload.

CloudFormation / CDK
Infrastructure as code. Define your stack in YAML/JSON or actual Python/TypeScript.

EBS — Block Storage
Disks for EC2. SSD/HDD options. Snapshots to S3.

Redshift — Data Warehouse
Petabyte-scale columnar SQL. Built on PostgreSQL. Fast for analytical queries.

Athena — Query S3 Directly
Serverless SQL over Parquet/CSV/JSON in S3. Pay per TB scanned.

SageMaker — ML Platform
Train, deploy, monitor ML models. Notebooks, endpoints, AutoML, feature store.

Bedrock — Foundation Models
Hosted Claude, Llama, Titan. API-only access to leading LLMs.

KMS — Key Management
Encrypt at rest in S3, EBS, RDS. Customer-managed keys for compliance.

Cloud Security: The Shared Responsibility Model

The fundamental rule: AWS secures **the cloud**. You secure **what you put in it**.

AWS handles	You handle
Hardware, network, datacenters	Your data, IAM policies
Hypervisor isolation	OS patching, app code
Managed-service infra	Your config, secrets

Famous breaches are almost always customer-side misconfig:

- **Capital One (2019)** – bad WAF → 100M records
- **Pegasus Airlines (2022)** – public S3 → 6.5 TB
- **Verizon (2017)** – public S3 → 14M records

5 pillars to defend:

1. **Identity** – IAM least privilege, MFA, never long-lived keys
2. **Network** – VPC, security groups, no `0.0.0.0/0`
3. **Data** – KMS at rest + TLS in transit, block public S3
4. **Detection** – CloudTrail, GuardDuty, Config, Security Hub
5. **Response** – runbooks, blast-radius isolation

> **#1 mistake**: over-permissive IAM role on a compromised EC2 instance.

AWS Shared Responsibility Model. https://aws.amazon.com/compliance/shared-responsibility-model/

Cloud Security in Action: Wiz

Wiz = cloud-native security platform (CNAPP). Agentless: it scans your cloud account via APIs — no software to install in your VMs.

The growth story:

- Founded **2020** by Assaf Rappaport, Ami Luttwak, Yinon Costica, Roy Reznik (ex-Microsoft, Adallom)
- Fastest SaaS ever to **$100M ARR** (18 months), **$500M ARR** (~3 years)
- **Google announced $32B acquisition** in March 2025 — Google's largest ever
- Used by **~50% of the Fortune 100**

Famous discoveries: ChaosDB (Azure Cosmos DB), OMIGOD, ExtraReplica.

The Wiz approach — the "Security Graph":

1. **Connect** to AWS / Azure / GCP via read-only API role
2. **Build a graph** of every resource, identity, network path, secret
3. **Trace attack paths** end-to-end: "this public S3 bucket → has IAM role → can read this DB"
4. **Prioritize** by actual blast radius, not raw CVE counts

Maps to the 5 security pillars:

- **Identity** — IAM analysis, over-permissive roles
- **Network** — exposed endpoints
- **Data** — sensitive data in public buckets
- **Detection** — CSPM, CWPP, KSPM, DSPM combined

> The lesson: **cloud security is a graph problem**, not a list of CVEs.

The Human Layer: Wombat & Phishing Training

Wombat Security Technologies = pioneer of **embedded phishing training**.

- Spun out of **CMU CyLab** in 2008 — founders: Norman Sadeh, Jason Hong, Lorrie Cranor
- Acquired by **Proofpoint in 2018 for $225M**

The premise: technical controls (Wiz, GuardDuty, WAF) don't matter if **one employee clicks a phishing link** and hands over credentials.

> **>90% of breaches start with a phishing email.** The weakest link in the cloud is human.

The PhishGuru method — "teachable moment":

1. Send employees a **simulated phishing email** (fake invoice, fake password reset, fake DocuSign)
2. If they click, **don't punish them** — show an immediate training page
3. The page highlights the **red flags** they missed: sender, URL, urgency, mismatched domain
4. Employee learns at the **moment of failure**, not in an annual slide deck

Result: published CMU studies show click-through rates drop **50–90%** after a few simulations.

> Phishing is a **conditioned reflex** problem. You retrain reflexes by **exposing people to fakes**, not by lecturing.

Kumaraguru, P., et al. Protecting people from phishing: the design and evaluation of an embedded training email system. CHI 2007.

Recap: Cloud vs. On-Premise

Cloud computing = compute, storage, memory, networking virtualized and exist on remote servers, rented by users.

Main pros of cloud vs. on-premise clusters:

1. **Manageability** — managing hardware is no longer the user's problem
2. **Pay-as-you-go** — fine-grained pricing economics based on actual usage (granularity: **seconds to years!**)
3. **Elasticity** — dynamically add or reduce capacity based on actual workload demand

Three layers: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS)

Review Questions

1. What are the **3 main layers** of a typical cloud? Give examples of AWS services in each layer. Which ones do your PAs use?
2. What is a **benefit of separating PaaS from SaaS** in cloud?
3. Briefly explain **1 pro and 1 con** of On-Demand vs Spot instances on AWS.
4. **What is so great about the serverless cloud anyway?**
5. Briefly explain **2 pros and 2 cons** of cloud vs. on-premise clusters.